

Import Data using Transform Maps (Spreadsheet)

REQUIREMENT ANALYSIS PHASE

1. Functional Requirements

- **Data Ingestion:**
 - The system must accept bulk employee data uploads from external spreadsheet formats (Microsoft Excel / .xlsx).
 - Data must first load into an intermediate staging/import set table (u_employee_import) before moving to production.
- **Data Schema & Storage:**
 - A custom target table (u_employee_test / Employee Test) must be created with specific attributes: Employee ID, Employee Name, Email, Department, and Location.
- **Transformation & Mapping:**
 - Administrators must be able to configure a Transform Map (Sample Spreadsheet Import) to connect source fields from the staging table to the correct target table fields using Auto-Mapping or Mapping Assist.
- **De-duplication Logic (Coalesce):**
 - The system must use **Coalesce** on the Employee ID field to evaluate incoming records.
 - If an Employee ID already exists, the record must be updated with new values rather than creating a duplicate entry. If the ID is new, a new record must be inserted.
- **Reporting & Visualization:**
 - The platform must generate custom reports based on the Employee Test table, including a Pie Chart for Department distribution, a Bar Chart for Location distribution, and a detailed List Report.
- **Dashboard Consolidation:**
 - Reports must be centralized into a custom dashboard (Employee Analytics Dashboard) to give HR and administrators a single-pane-of-glass view.

2. Non-Functional Requirements

- **Data Integrity & Accuracy:** Bulk imports must maintain zero unintended data loss, accurately tracking inserts, updates, and ignored/skipped rows during re-imports.
- **Performance & Efficiency:** Batch data transformations must execute swiftly (e.g., processing 15+ records within seconds) without system degradation.
- **Security & Access Control:**
 - Access controls (ACLs) must be properly configured (e.g., enabling Admin Overrides on pa_dashboards) to allow administrators to build and manage analytics dashboards safely.
 - Dashboards and reports must support role-based sharing restrictions to limit visibility to authorized groups and users.

3. Assumptions & Constraints

- **Assumptions:**
 - External data is provided cleanly in a tabular Excel format with standard headers on row 1.
 - The administrator executing the import has global application scope access and necessary table-creation privileges.
- **Constraints:**
 - Matching keys (such as Employee ID) must remain uniform between source files and target records to ensure the Coalesce feature works correctly.